Homework 3

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1. Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(flextable) # creating tables

# create kelp object with data
kelp <- read_csv(here("data", "temp-kelp.csv"))
# create bike_commute data with personal data
bike_commute <- read_xlsx(here
                          ("data","ENVS_193DS_Personal_Data_Project.xlsx"))
```

2. Problems

Problem 1. Giant kelp fronds

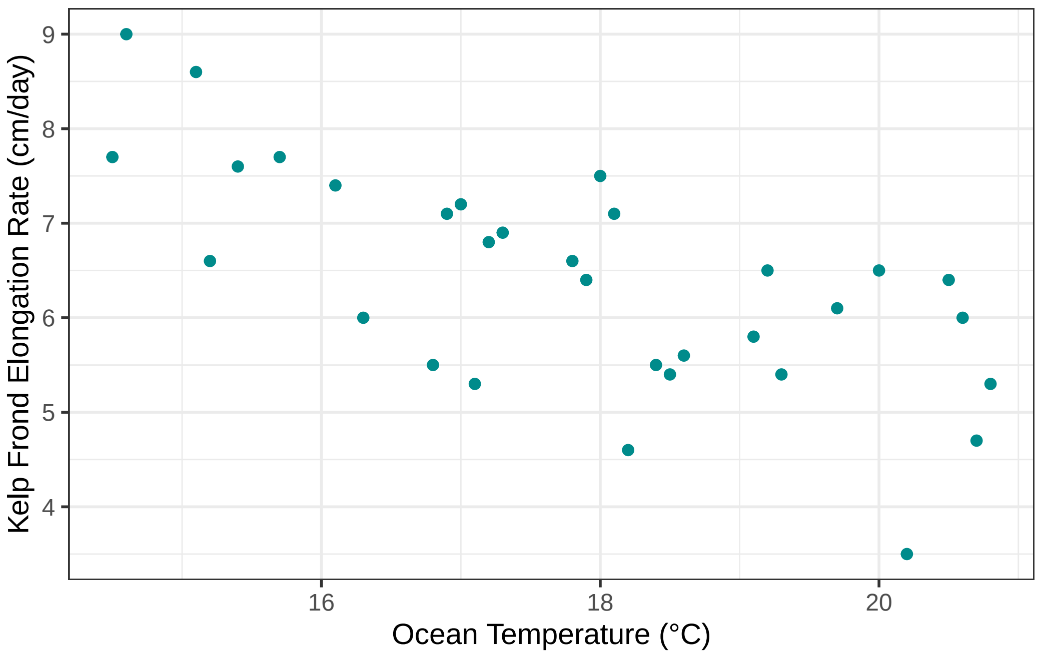
a. An appropriate test

Because we are interested in the strength of the relationship between temperature and giant kelp frond elongation rate, the appropriate test is a Pearson's r or a Spearman rank correlation. These tests both test for correlation between two variables and see haw the variables move together, not necessarily predict each other. Pearson's r test is a parametric test that can be used for linear relationships that have continuous and normally distributed variables. Spearman rank correlation is a non-parametric test, which can be used for monotonic relationships that have interval, ordinal, or nominal data.

b. Create a visualization

```
# base layer: ggplot
# using kelp data frame
ggplot(data = kelp,
       mapping = aes(
          x = temp_c, # x- axis: Ocean temperature (degrees C)
          y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
  # first layer: points
  geom_point(
```

```
    #changing color from default
    color = "darkcyan") +
  # relabeling x- and y- axes
  labs(
    x = "Ocean Temperature (°C)", #
    y = "Kelp Frond Elongation Rate (cm/day)") +
  # changing theme from default
  theme_bw()
```

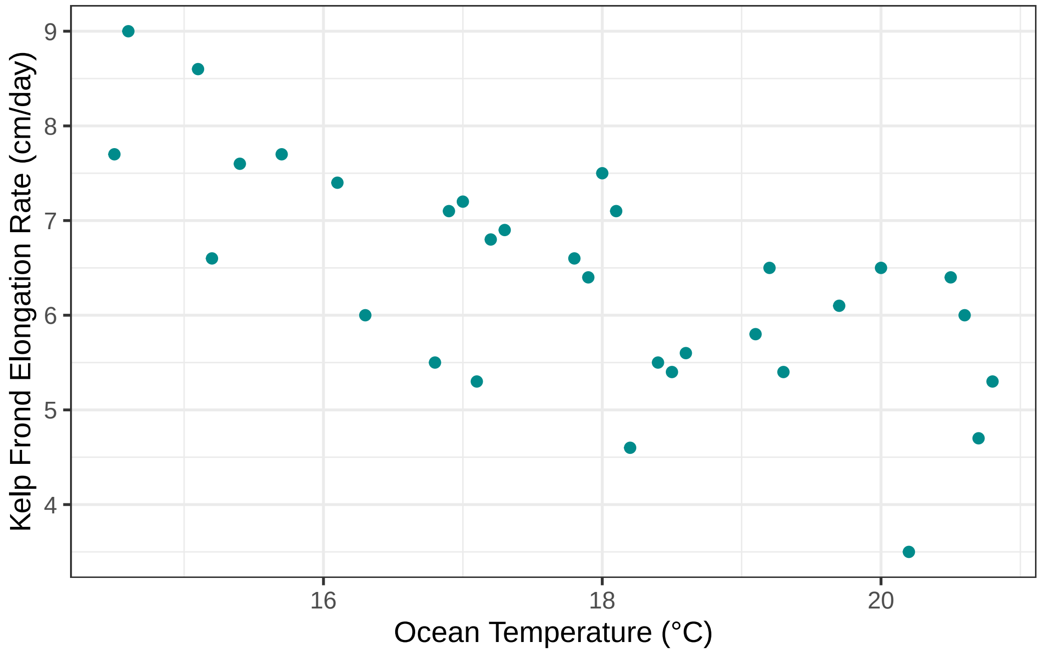


c. Check your assumptions and run your test

Linearity

```
# base layer: ggplot
# using kelp data frame
ggplot(data = kelp,
       mapping = aes(
          x = temp_c, # x- axis: Ocean temperature (degrees C)
          y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
  # first layer: points
  geom_point(
```

```r
    #changing color from default
    color = "darkcyan") +
  # relabeling x- and y- axes
  labs(
    x = "Ocean Temperature (°C)", #
    y = "Kelp Frond Elongation Rate (cm/day)") +
  # changing theme from default
  theme_bw()
```

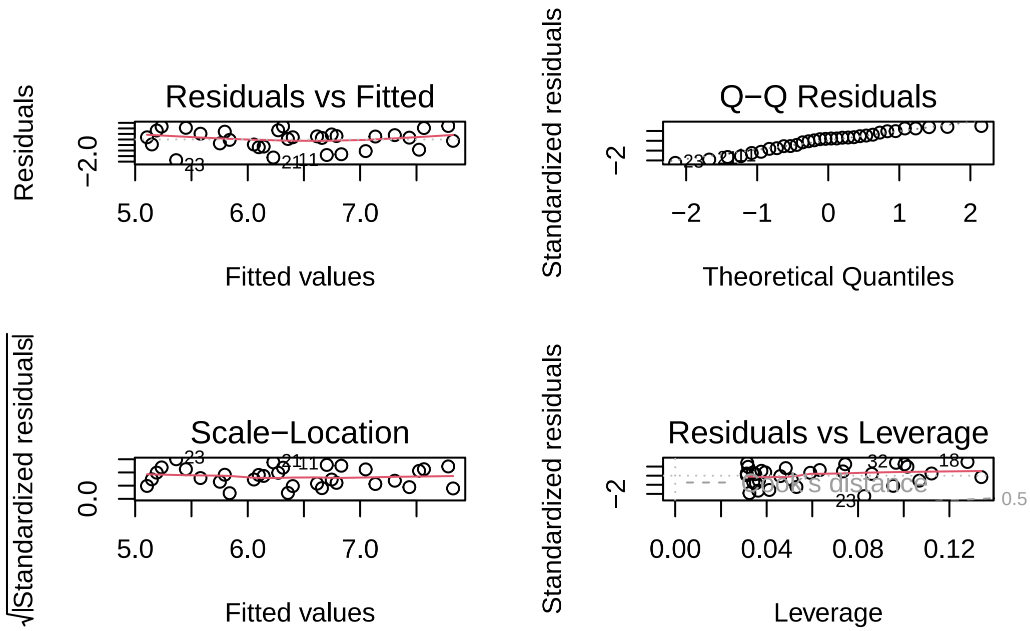


Model fitting

```r
kelp_model <- lm(
  kelp_elong ~ temp_c, # formula: change in kelp length as a function of temp
  data = kelp)       # data: kelp data
```

Diagnostics

```r
par(mfrow = c(2, 2))   # displays plots in a 2 by 2 grid
plot(kelp_model)     # displays diagnostic plots for model
```



The data is linear and continuous meaning a pearsons r test should be used. In the diagnostic plots, the Q-Q Residuals plot shows that the residuals are normally distributed. The Residuals vs Fitted and Scale-Location plots show that the residuals are evenly distributed with no clear patterns. And finally, the Residuals vs Leverage plot shows that their are no outliers in the data that may have a large influence on the models.

Running test

```
# running correlation test
# variables ocean temperature and giant kelp elongation rate
cor.test(kelp$temp_c, kelp$kelp_elong,
    # running pearson's r based on assumptions
    method = "pearson")
```

```

    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
```

```
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a

To evaluate the strength of the relationship between ocean temperature and giant kelp (Macrocystis pyrifera) frond elongation rate, I used a Pearson's r correlation test. The test revealed a moderate to strong negative (Pearson's r = -0.69) relationship between ocean temperature and kelp elongation rate(cm/day) ($t(30) = -5.19$, $p < 0.001$, $\alpha = 0.05$).

e. Test implications

Our results revealed a strong to moderate negative relationship between ocean temperature and Giant Kelp Elongation rate. This means that at higher ocean temperatures, Giant Kelp frond elongation rate is lower and therefore, to maximize Giant Kelp growth, ocean water temperature should be at a natural low. As average ocean temperature rises due to climate change, we will most likely see a decrease in Giant Kelp Elongation rate.

f. Double check your own work.

```r
# running correlation test
# variables ocean temperature and giant kelp elongation rate
cor.test(kelp$temp_c, kelp$kelp_elong,
    # Spearman's rank test
    method = "spearman")
```

```
Warning in cor.test.default(kelp$temp_c, kelp$kelp_elong, method = "spearman"):
Cannot compute exact p-value with ties


    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
```

```
sample estimates:
       rho
-0.6891684
```

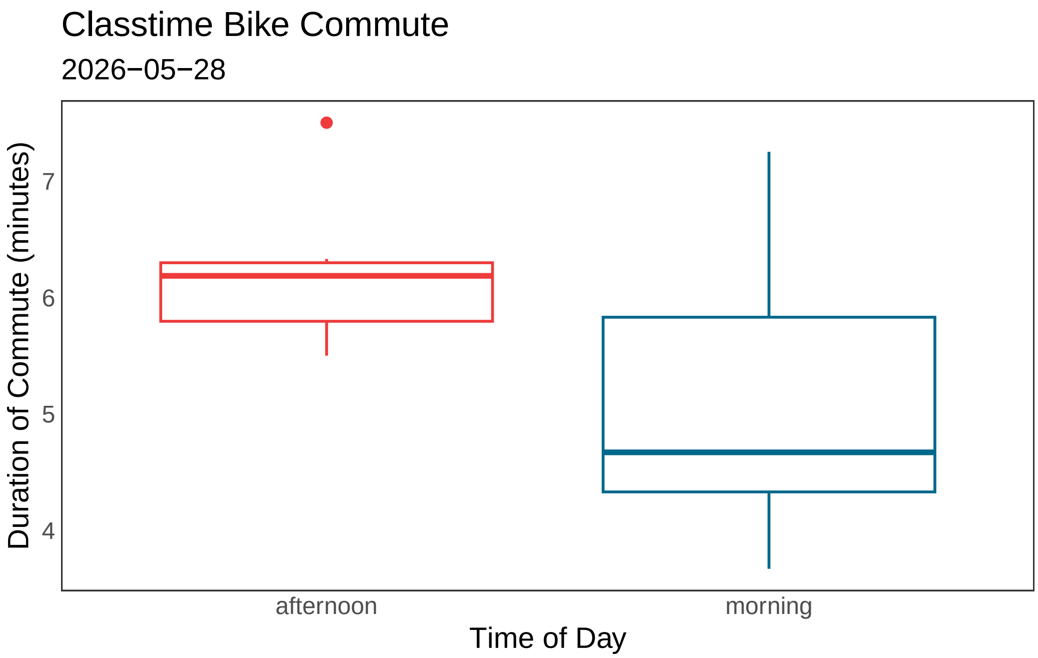
The results from the Spearman's rank test illustrate that I would have made the same conclusions that I did with the Pearson's test. The Spearman's rank test also gave a p-value < 0.001 and gave a correlation coefficient that is -0.69.

Problem 2. Personal Data

a. Updating your visualizations

```
# base layer: ggplot
# data frame: bike_commute
ggplot (data = bike_commute,
        mapping = aes(
          # x-axis: time of day
          # y-axis: duration of commute
          x = time_of_day,
          y = `duration_(m)`,
          # color data by time of day
          color = time_of_day
        )) +
  # first layer: boxplot
   geom_boxplot() +
  # Changed title, x and y-axis
  labs (title = "Classtime Bike Commute",
        x = "Time of Day",
        y = "Duration of Commute (minutes)",
        subtitle = "2026-05-28") +
  # changed the color of data to custom
  scale_color_manual(
    values = c(
      "afternoon" = "brown2",
      "morning" = "deepskyblue4"
    )) +
  # applied custom plot theme
  theme_bw() +
  # removing the panel background
  theme(panel.background = element_blank()) +
```

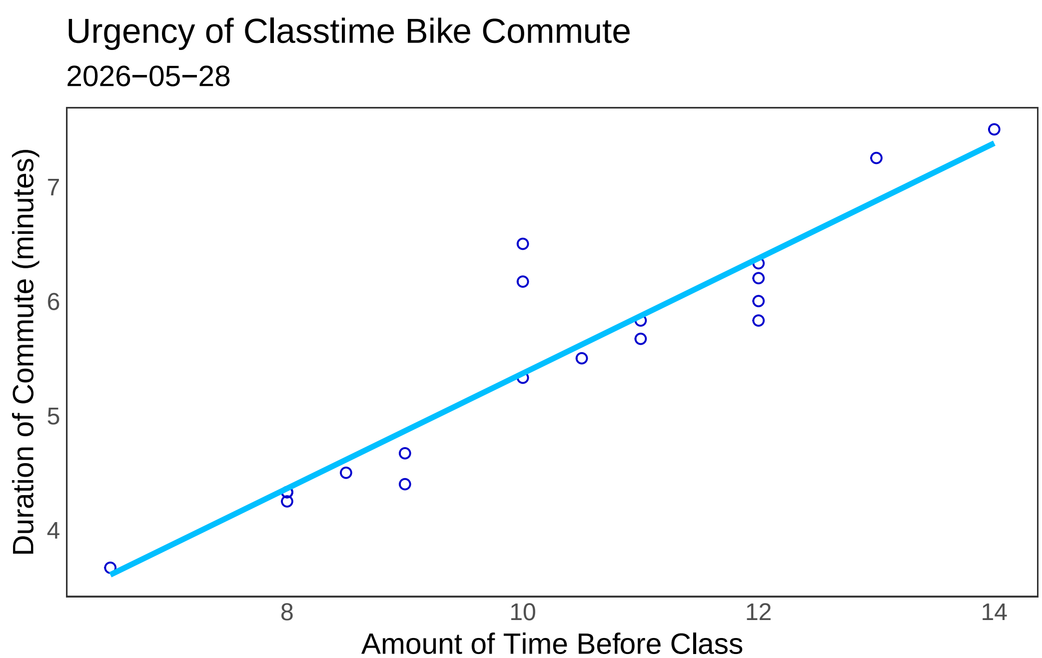
```
  # removing the panel grid
  theme(panel.grid = element_blank()) +
  # removing axis ticks
  theme(axis.ticks = element_blank()) +
  # removed legend
  theme(legend.position = "none")
```




```
# base layer: ggplot
# data frame: bike_commute
ggplot (data = bike_commute,
        mapping = aes(
          # x-axis: amount of time before class
          # y-axis: duration of commute
          x = `time_before_class_(min)`,
          y = `duration_(m)`
        )) +
  # first layer: point = scatterplot
  # make data points colored, change shape
  geom_point(color = "blue3",
             shape = 21) +
  # add reference linear line that fits model, make line a lighter blue
  geom_smooth(method = "lm", se = FALSE,
```

```
              color = "deepskyblue") +
  # Changed title, x and y-axis
  labs (title = "Urgency of Classtime Bike Commute",
        x = "Amount of Time Before Class",
        y = "Duration of Commute (minutes)",
        subtitle = "2026-05-28") +
  # applied custom plot theme
  theme_bw() +
  # removing the panel background
  theme(panel.background = element_blank()) +
  # removing the panel grid
  theme(panel.grid = element_blank()) +
  # removing axis ticks
  theme(axis.ticks = element_blank())
```

```
`geom_smooth()` using formula = 'y ~ x'
```



b. Captions

Figure 1. Morning bike commute duration on average have a shorter commute time than afternoon bike commute duration. Red box plot represents observations of

afternoon bike commute duration in minutes and blue box plot represents observations of morning bike commute duration in minutes. The center line displays the median of observations with box and whiskers representing upper and lower quartiles. Data is from ENVS 193DS personal data collection project.

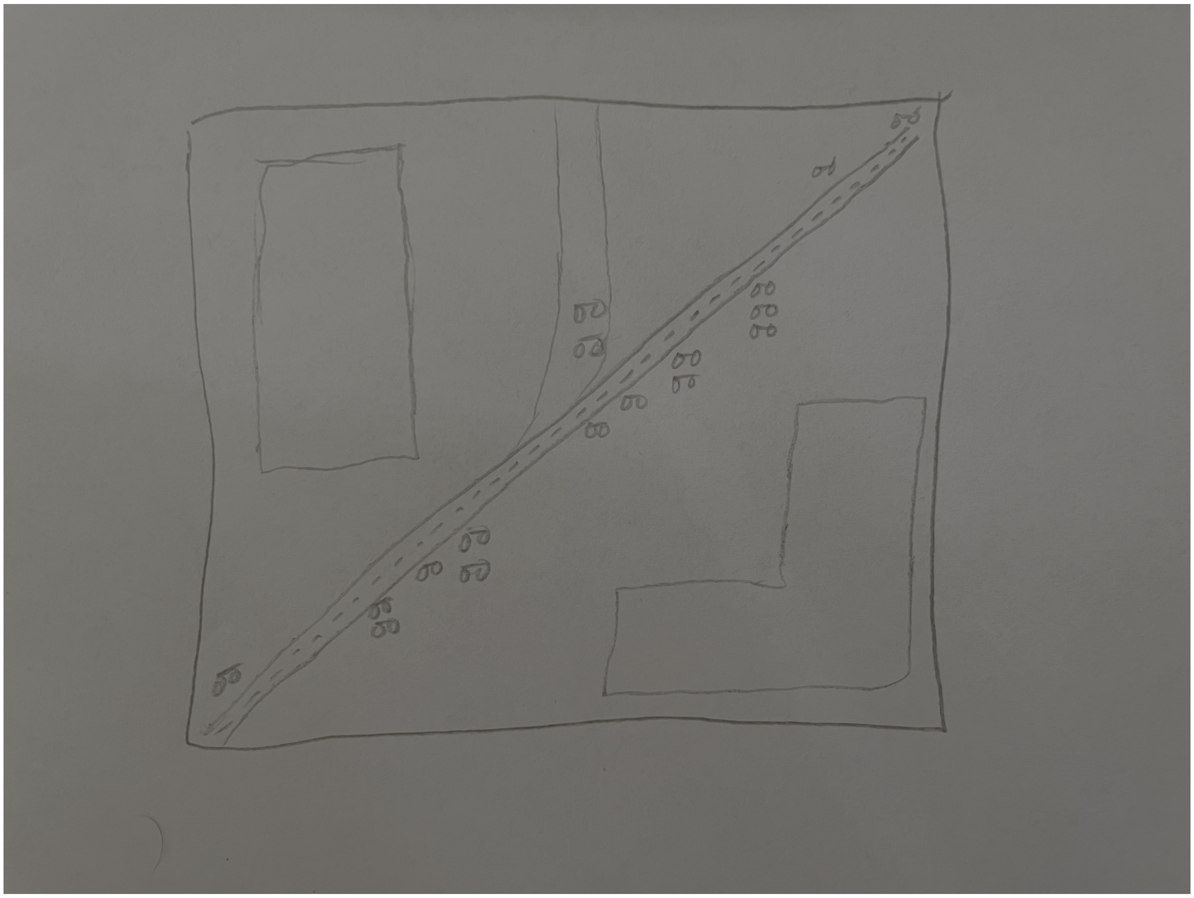
Figure 2. Commuting to class with less time before it starts, correlates to a shorter duration of commute time. Points represent individual observations with "Amount of Time Before Class" on the x-axis and "Duration of Commute (minutes)" on the y-axis. Data is from ENVS 193DS personal data collection project.

Problem 3. Affective visualization

a. Describe in words what an affective visualization could look like for your personal data

I think that my personal data showing duration of bike commute as it relates to the amount of time I leave home before class shows an interesting linear relationship. I think this is an important relationship to visulaize and could be done in a creative way for my affective visualization to effectively explain the data. I could use the shape of the linear line to make a bikepath running through campus with individual points as specific objects (little bikers or trees).

b.Create a sketch (on paper) of your idea.



c. Make a draft of your visualization.

d. Write an artist statement.

- the content of your piece (what are you showing?)

  The visualization reflects data of a linear relationship of bike commute relationship compared to amount of time available before class starts.

- the influences (what did techniques/artists/etc. did you find influential in creating your work?)

  The piece is influenced by the bike culture at UCSB and the necessity to efficiency bike to class.

- the form of your work (watercolor, oil painting, crocheted object, etc.)

  This piece is digital artwork.

- your process (how did you create your work?)

  This piece was made on Canva using the platforms digital artwork resources.

e. Prep your materials to share in class.

Link to Slides presentation here

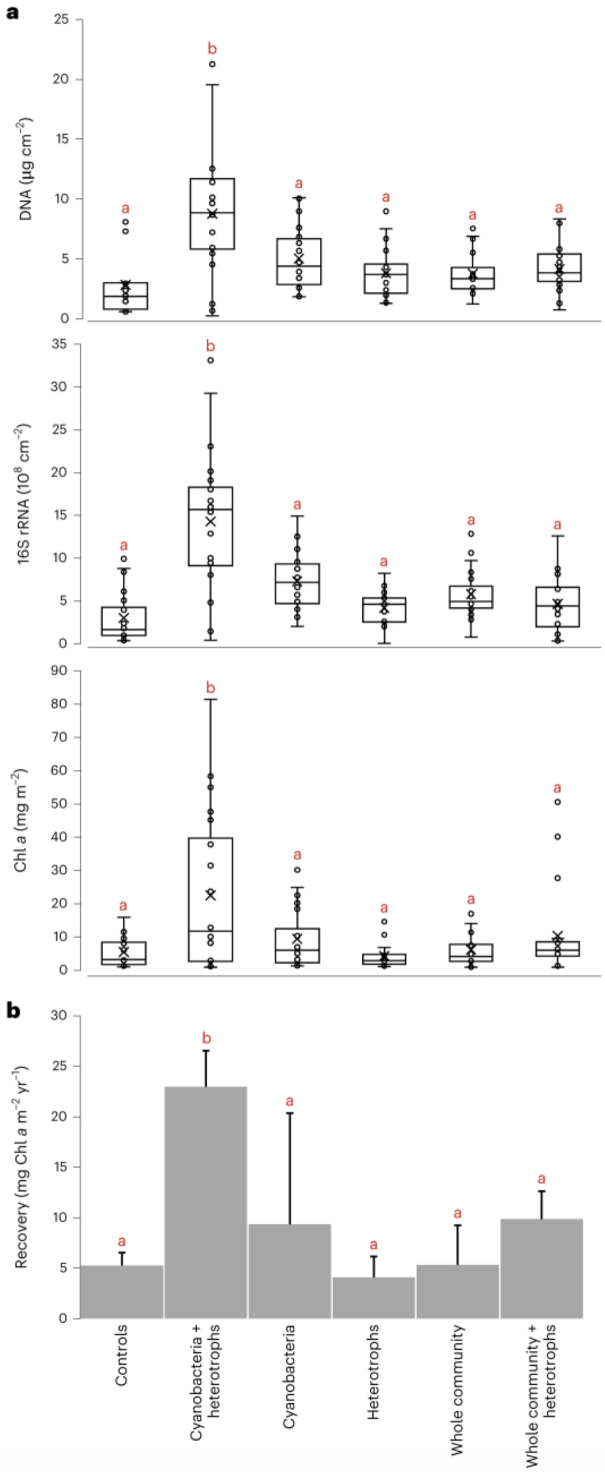
Problem 4. Statistical critique

a. Revisit and summarize

This research paper uses a Kruskal-Wallace test. The response variable of this test is biocrust characteristics and related measurements such as biocrust cover, biomass, and chlorophyll content. The predictor variable is the treatment area the data was collected from, such as, under solar panels or open/exposed areas as well as what type of inoculum treatment was added to the soil to stimulate biocrust.

Fig. 5: Effect of inoculum type on end-point biocrust development and recovery rates.

From: Dual use of solar power plants as biocrust nurseries for large-scale arid soil restoration



b. Visual clarity

The above visualization includes 4 sepearte plots with clear x and y-axis. The y-axis shows the categorical data and the x-axis of each plot represents a different response variable. Underlying data is shown as well as summary statistics from boxplots (medians upper and lower quartiles).

c. Aesthetic clarity

In my opinion the data:ink ratio is creating a little bit of visual clutter. Displaying 4 graphs with box and whisker plots all with 6 categorical variables creates clutter. The plots could have been separated/organized to be more concise.

d. Recommendations

The plots could have been separated/organized to be more concise into related representations instead of including all 4 plots in one figure. Also if the data allows it, a simple mean and standard error could represent the data with less clutter.